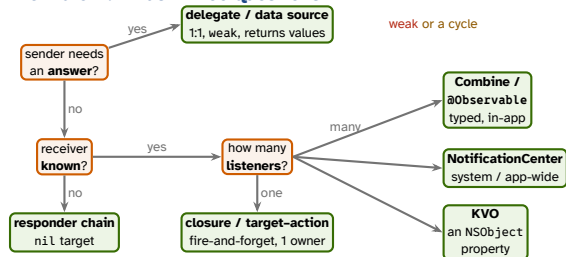
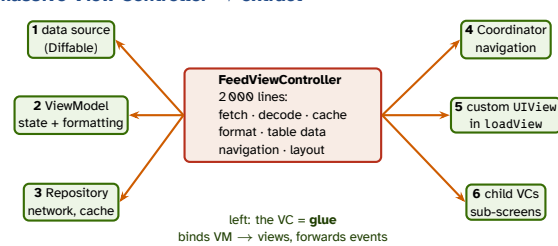


In one line: Cocoa gives you a small set of **communication idioms** — delegate, data source, target-action, closure, NotificationCenter, KVO, responder chain — and choosing one is a question of **how many listeners**, **whether the sender needs an answer**, and **who owns whom**. Most iOS anti-patterns are one mistake: **responsibilities and dependencies that nobody can see**.

Which idiom? — ask three questions



Massive View Controller → extract



The idioms

Idiom	Shape	Mechanism · rule
Delegate	1:1	protocol : AnyObject, weak var delegate; can answer (shouldSelect), did/will/should names, sender first delegate that <i>supplies data</i> (numberOfRows); UIKit holds both weak
Data source	1:1	
Target-action	1:N	addTarget(_:action:for:) + @objc selector; the control does not retain the target; UIAction closure (iOS 14)
Closure	1:1	onSelect: (Item) → Void; stored → [weak self]; best for one event
Notification-Center	1:N	broadcast by name, untyped userInfo; synchronous, on the posting thread
KVO	1:N	observe(\.x, options:) → NSKeyValueObservation; needs NSObject + @objc dynamic
Responder chain	1:?	sendAction(_:to: nil, _): first responder up via next (see gof-behavioural)
Combine / @Observable	1:N	typed streams / per-property tracking (iOS 17); the modern default in-app

```
protocol PickerDelegate: AnyObject { // so it can be weak
    func picker(_ p: Picker, didPick item: String)
    func pickerShouldClose(_ p: Picker) -> Bool // answer back
}

final class Picker {
    weak var delegate: PickerDelegate? // Picker never owns it
    func pick(_ item: String) {
        delegate?.picker(self, didPick: item)
        if delegate?.pickerShouldClose(self) ?? true { /* close */ }
    }
}

// load(true, false) -- what? Name the choice instead:
enum CachePolicy { case useCache, reload }
func load(_ policy: CachePolicy) {} // load(.reload)
```

Massive VC — the recipe

Characterisation/snapshot tests first, then steps 1–6 as drawn, one safe move at a time (child VCs: addChild + didMove(toParent:)). It is an **SRP** problem, not an MVC one — moving the mass into the model or a Manager only relocates it.

Traps

- ! “Singletons are always bad” — no (FileManager.default).
- The damage is **hidden lookup** (.shared inside logic) + **global mutable state**: keep one instance, **inject** it.
- ! A service locator still hides dependencies — a missing registration fails at run time, not compile time (Mark Seemann).
- ! NotificationCenter for a 1:1 flow = invisible coupling: no contract, no “who listens?”.
- ! “Duplication is far cheaper than the wrong abstraction” (Sandi Metz).
- ! UIAction/sink/timer closures are *stored*: [weak self]; unowned only when the lifetime is guaranteed.

Remember

Answer back → **delegate** · **one** → **closure** · **many** → **Combine/Notification** · **unknown** → **responder chain**. Make responsibilities and dependencies visible.

Anti-pattern	Symptom	Damage	Fix
Massive VC	1000+ lines, everything in viewDidLoad	untestable, merge conflicts	the recipe above; VC becomes glue
God object	AppManager, Utils; everyone imports it	every change ripples	split by responsibility; inject small collaborators
Singleton / locator	.shared / resolve() inside logic	hidden deps, order-dependent tests	init injection; .shared only at the composition root
Stringly-typed	segue ids, userInfo["k"], [String: Any]	typos crash at run time	enums, Notification.Name constants, Codable
Boolean flags	load(true, false)	unreadable calls, 2 ⁷³ branches	two functions, an enum, or an OptionSet
Deep inheritance	BaseViewController with 30 hooks	fragile base class, LSP breaks	protocols + extensions, composition, child VCs
Copy-paste reuse	the same fix needed in 5 places	shotgun surgery, drift	extract once it repeats 3 ×
Premature abstraction	a protocol with one conformer	indirection, harder to read	YAGNI; abstract at the 2nd–3rd case (a test seam counts)
Callback pyramid	nested completions; a path never calls back	lost errors, hangs	async/await, structured concurrency
UIKit in models	UIColor/UIImage in domain types	no widget/test reuse, main-thread ties	Foundation-only domain; map in the ViewModel
Over-modularization	60 modules, one file each	build graph + API churn cost	module by feature/team, measure build times

Related: gof-behavioural (responder chain, Observer) · observation-and-combine · arc-ownership · solid · mvc-mvp-mvvm · coordinator-repository-di-clean · testable-design-seams · modularization-spm